Block to Python Bridge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Block to Python Bridge before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Requests Responses and JSON

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Conversation State and Memory

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Model Memory vs Retrieval

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Block to Python Bridge trace and record the first step you can explain without help.

Use W2: Predict then observe for From a paper **algorithm** to Python. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Block to Python Bridge explanation against the supplied facts before discussing it.

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

> Set total=0, then for values2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Block to Python Bridge target to a trace and a justified conclusion.

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Block to Python Bridge case and show the deciding evidence.

A block **program** sets total=0 and adds each value in [2,4,6].

Translate and trace.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Trace this Python: score = 2; then if score >= 2: score = score + 3. What is the final value, and what changes if the **condition** becomes score > 2?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 23 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Block to Python Bridge adds to Verified Study Buddy and keep the evidence.

Implement small transparent **function**s for selection, validation, and citation formatting.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Connect the paper steps to Python

> Connect the paper steps to Python. Apply the supplied Block to Python Bridge facts and keep your reasoning.

Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Predict then observe

> Predict then observe. Apply the supplied Block to Python Bridge facts and keep your reasoning.

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Explain an order error

> Explain an order error. Apply the supplied Block to Python Bridge facts and keep your reasoning.

Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the procedure independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the procedure independently. Apply the supplied Block to Python Bridge facts and keep your reasoning.

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Keep an honest conclusion

> Keep an honest conclusion. Apply the supplied Block to Python Bridge facts and keep your reasoning.

Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Block to Python Bridge

> Translate a “count qualifying books” paper procedure into a Python-style trace. Explain the labelled step before checking the changed condition.

Translate a “count qualifying books” paper procedure into a Python-style trace.

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/23

Block to Python Bridge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Start counter

count begins at 0 before the loop.

2. Trace decisions

80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.

3. Report count

The final result is 2 qualifying books, not the sum of their page counts.

CHANGE ONE THING

Replace the last page count 90 with 100.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Translate a "count qualifying books" paper procedure into a Python-style trace.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final result is 2 qualifying books, not the sum of their page counts.

1. Trace decisions

2. Start counter

3. Report count

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report count The final result is 2 qualifying books, not the sum of their page counts. Read the picture in order.

The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace the last page count 90 with 100. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The final count becomes 1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The final count becomes 1. The

> comparison is strictly less than 100, so the boundary value does not qualify.

Repair a missing building block

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

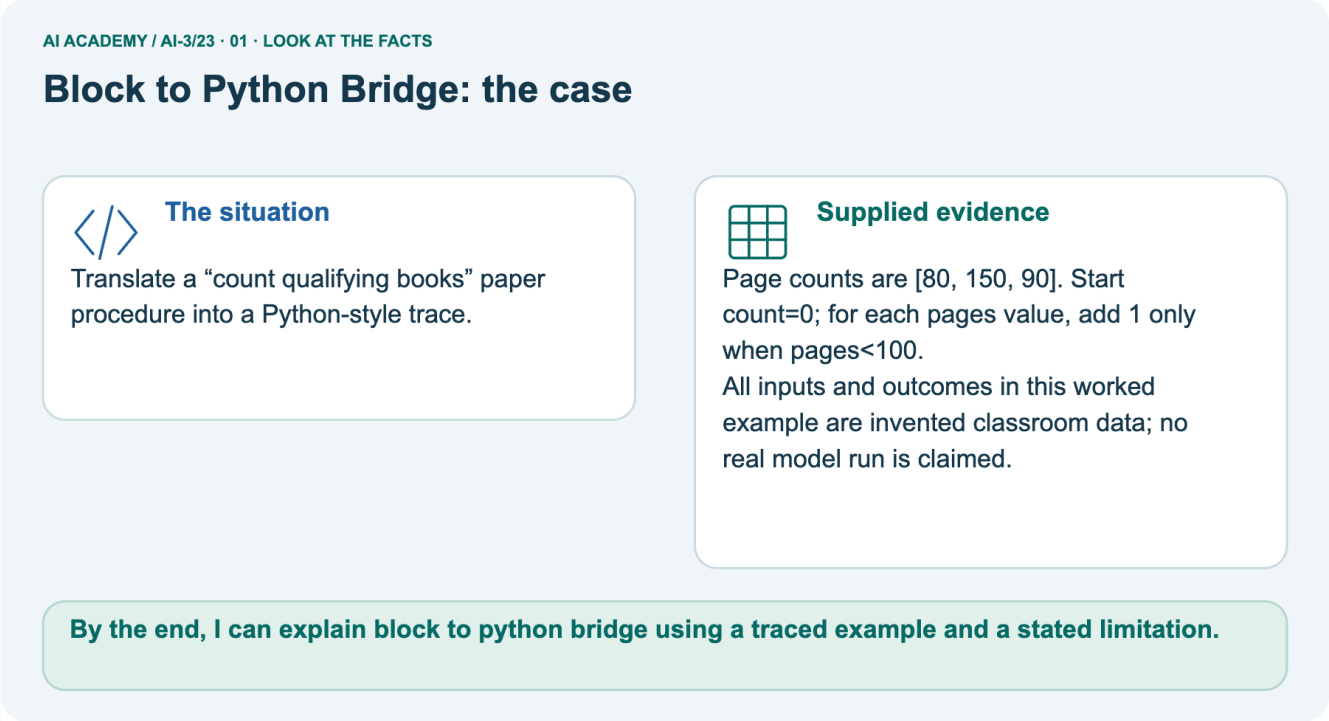
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Block to Python Bridge picture from its labels, then record a relevant limit.

01. Block to Python Bridge: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

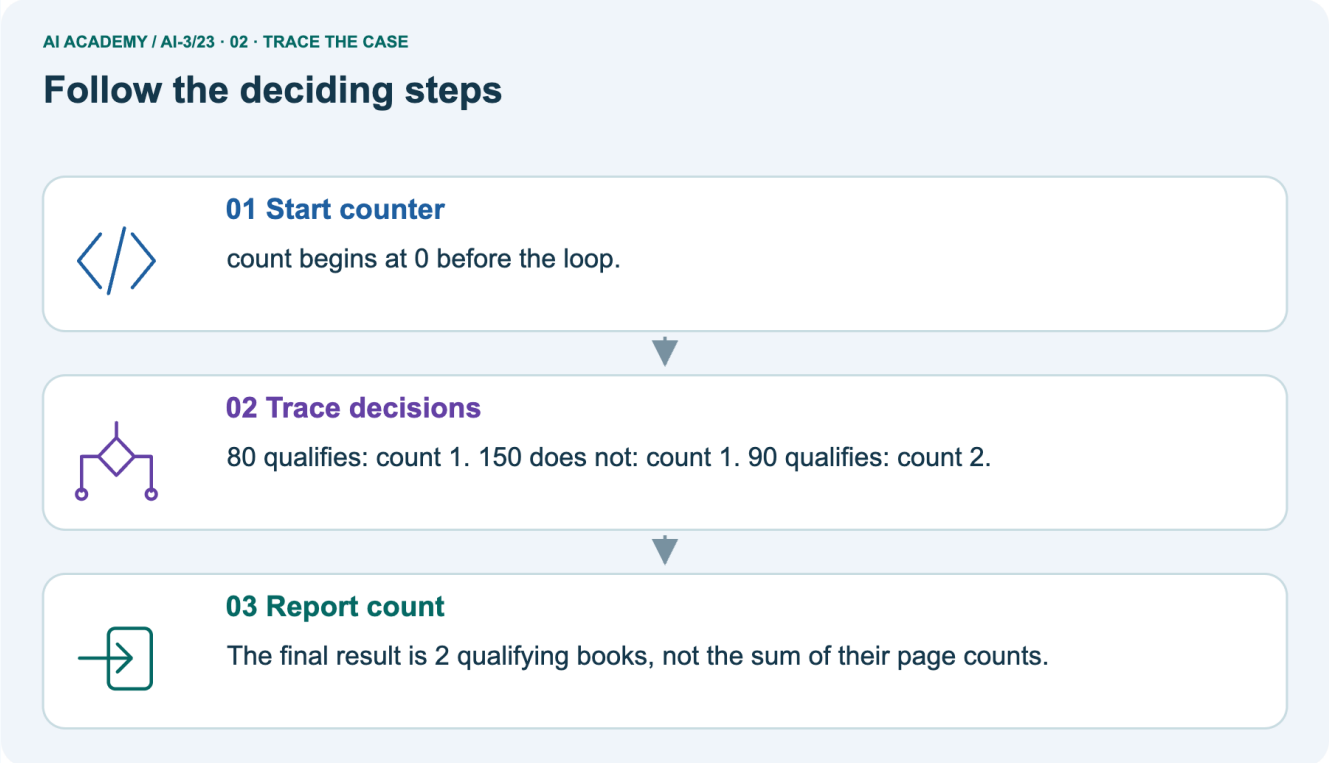
Translate a “count qualifying books” paper procedure into a Python-style trace. Supplied facts: Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



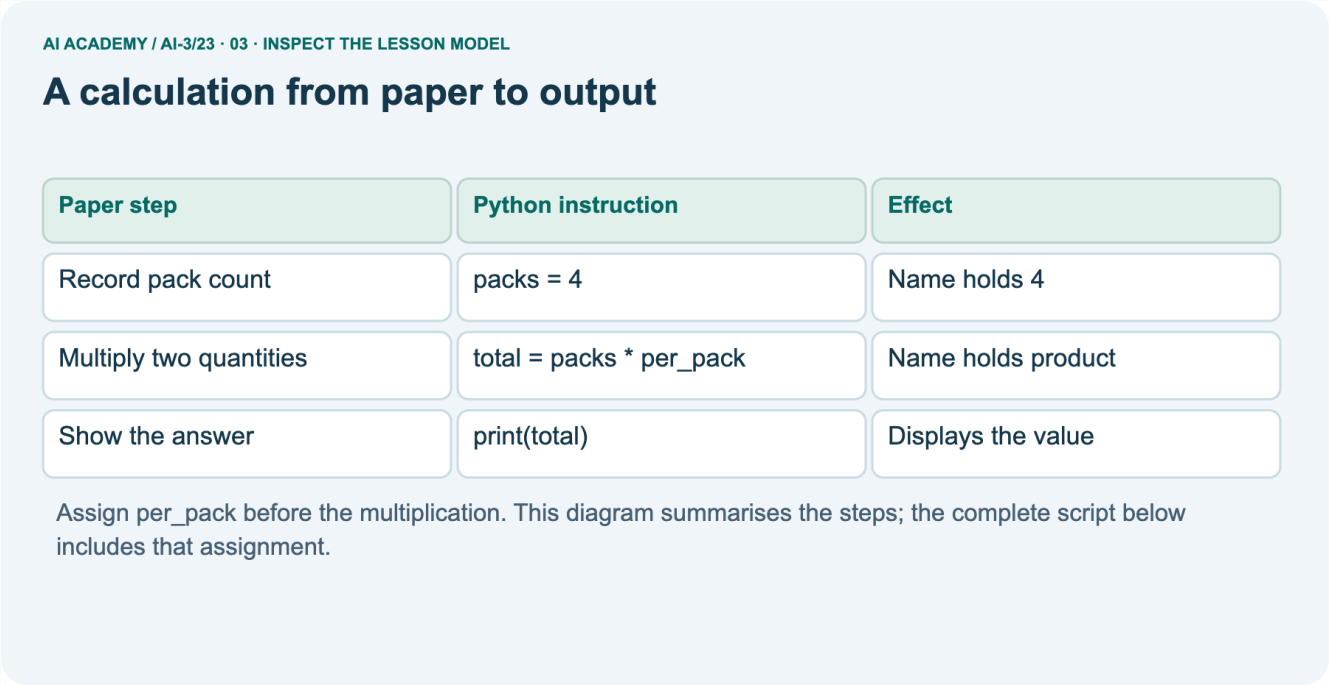
Follow the labelled picture with the matching explanation. Open the vector diagram

Start counter: count begins at 0 before the loop. → Trace decisions: 80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2. → Report count: The final result is 2 qualifying books, not the sum of their page counts.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A calculation from paper to output



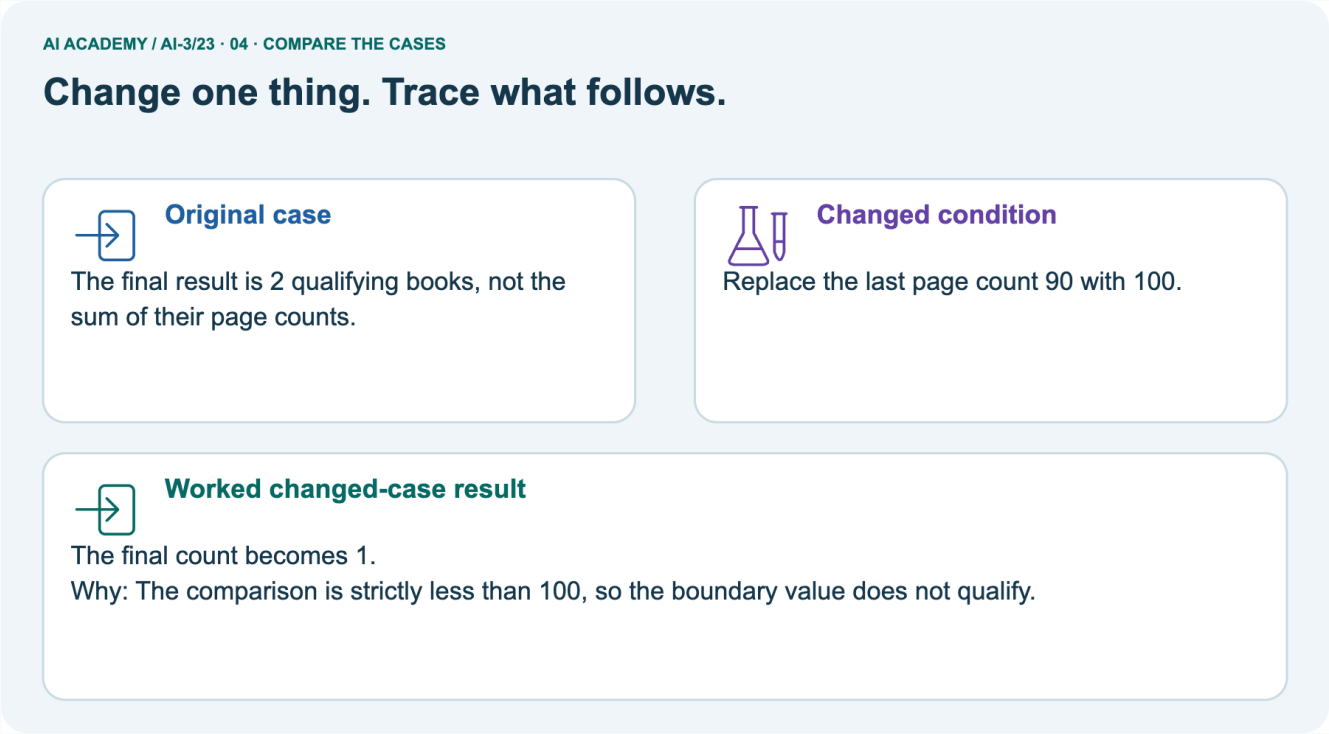
Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

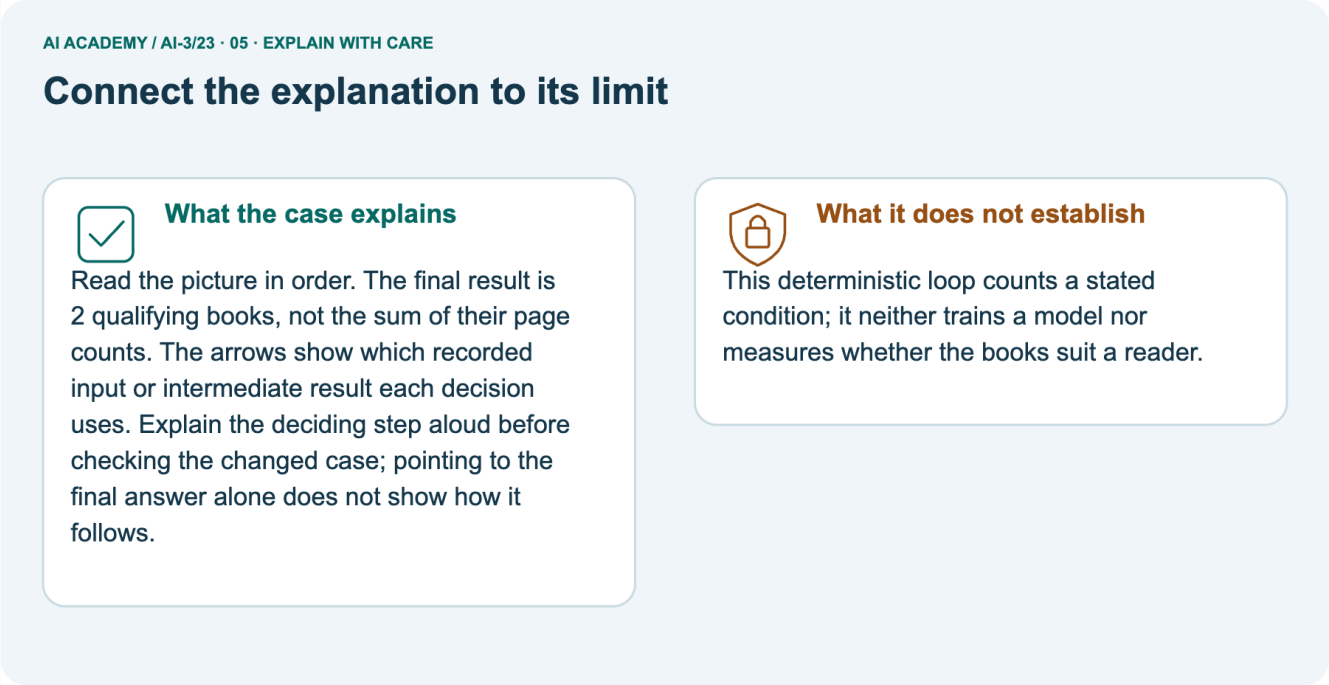
Original case: The final result is 2 qualifying books, not the sum of their page counts. Changed **condition**: Replace the last page count 90 with 100. Expected result: The final count becomes 1.

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

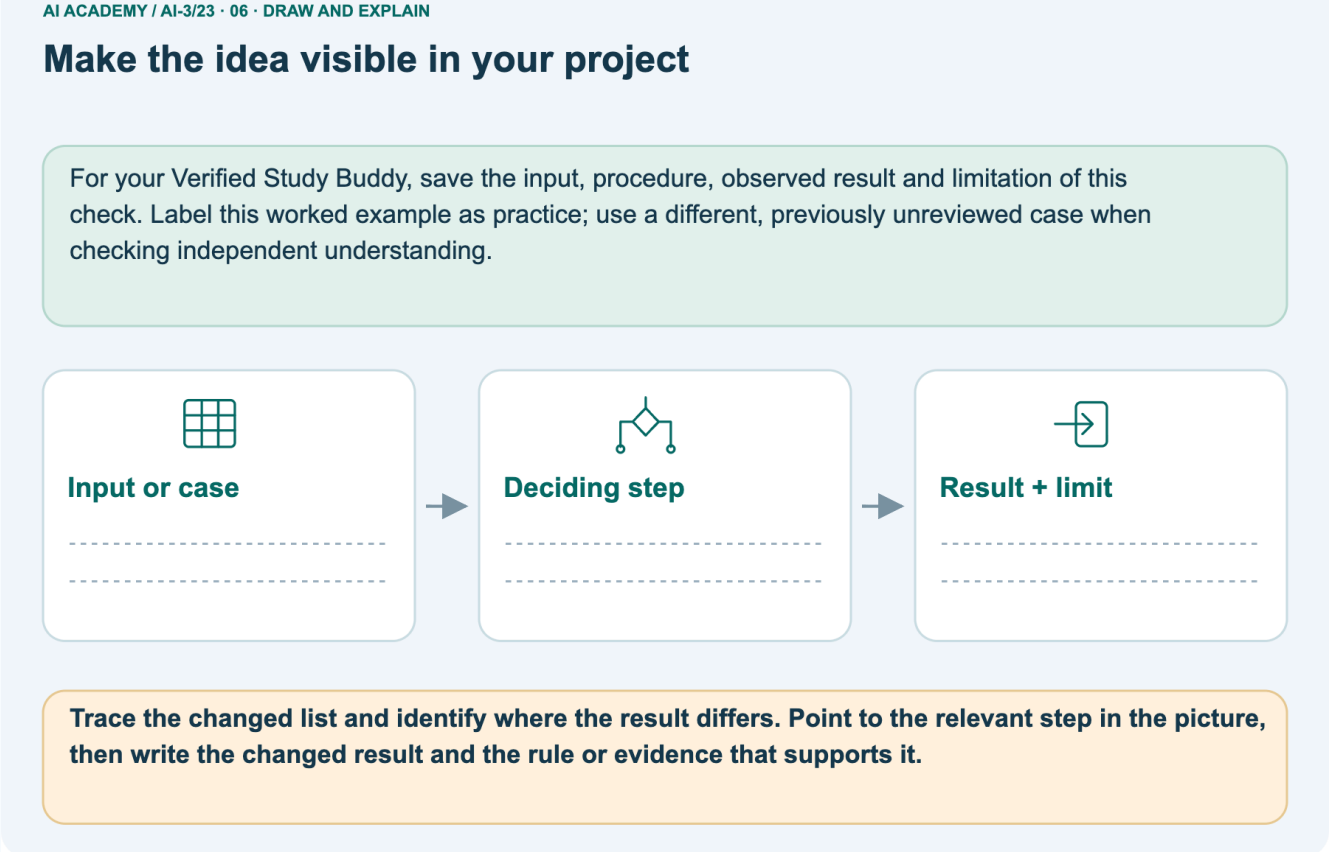
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.